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Title

Mechatronic Project 478
Final Report

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Student number	Signature
Initials and surname	Date

Executive summary

Title of Project
Joint Design for a Portable Coordinate Measurement Arm (PCMA)
Objectives
To derive technical performance measures (TPMs) from an error-propagation model, and use them to design, build, and test a position-holding joint that maintains and /or improves the arm's overall accuracy.
What is current practice and what are its limitations?
Current joints rely on counterbalancing and lack holding torque, meaning the operator must physically support the arm's weight during operation.
What is new in this project?
The introduction of a position-holding mechanism specifically engineered using mathematically derived TPMs.
If the project is successful, how will it make a difference?
It will directly maintain or improve the physical measurement accuracy of the arm. Additionally, it will yield a modified kinematic model to serve as a generalized framework for future accuracy analysis.
What are the risks to the project being a success? Why is it expected to be successful?
Risks: University workshop manufacturing capabilities may fall short of required tolerances, and the holding mechanism could introduce unwanted friction. Expectation of Success: The design requirements are heavily grounded in an established, proven error model.
What contributions have/will other students made/make?
The project builds directly upon an initial error-propagation model developed by Stellenbosch alumna Leon Hall.
Which aspects of the project will carry on after completion and why?
The modified error-propagation model. It will be able to accept custom arm dimensions and compute the uncertainty distribution within a given work volume, forming the mathematical foundation for future PCMA development.
What arrangements have been/will be made to expedite continuation?

Extensive, standardized documentation will be left behind to ensure future researchers can build directly onto this work without repeating groundwork.

Acknowledgements

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List of symbols

Constants

$L_0 =$	300 mm	

Variables

Re_{D}	Reynolds number (diameter)	[ ]
x	Coordinate	[m]
$\ddot{x}$	Acceleration	[m/s^2]
θ	Rotation angle	[rad]
τ	Moment	[N·m]

Vectors and Tensors

$\vec{v}$	Physical vector, see equation ...

Subscripts

a	Adiabatic
a	Coordinate

Abreviations

PCMA	portable coordinate measurement arm
FEA	Finite Element Analysis

Chapter 1

Introduction

1.1 Background

Starting from the big picture, gradually narrow focus down to this project and where this report fits in.

1.2 Objectives

The objectives of the project (in some cases the objectives of the report). If necessary describe limitations to the scope.

1.3 Motivation

Why this specific project/report is worthwhile.

Chapter 2

Literature review

2.1 Discrete element method

The Discrete Element Method (DEM) analysis (Cundall and Strack, 1979) uses spherical objects. Lin and Ng (1997) developed a DEM model for ellipsoids.

Chapter 3

Content chapter

Unless the chapter heading already makes it clear, an introductory paragraph that explains how this chapter contributes to the objectives of the report/project.

3.1 Heading level 2

3.1.1 Heading level 3

3.1.1.1 Deepest heading, only if you cannot do without it

Equations: An equation must read like part of the text. The solution of the quadratic equation $ax^2 + bx + c = 0$ given by the following expression (note the full stop after the equation to indicate the end of the sentence):

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2b}. \tag{3.1}$$

In other cases the equation is in the middle of the sentence. Then the paragraph following the equation should start with a small letter. Euler's identity is

$$e^{i\pi} + 1 = 0, \tag{3.2}$$

where e is Euler's number, the base of natural logarithms.

The `amsmath` has a wealth of structure and information on formatting of mathematical equations.

Symbols and numbers: Symbols that represent values of properties should be printed in italics, but SI units and names of functions (e.g. sin, cos and tan) must not be printed in italics. There must be a small hard space between a number and its unit, e.g. 120 km. Use the `siunitx` package to typeset numbers, angles and quantities with units:

`\num{1.23e3}`	→	1.23×10^3
`\ang{30}`	→	30°
`\qty{20}{N.m}`	→	20 N·m

Figures and tables: The `graphicx` package can import `PDF`, `PNG` and `JPG` graphic files.

Table 3.1: Standard ISO paper sizes

Paper	Sizes	
	W [mm]	H [mm]
A0	841	1189
A1	594	841
A2	420	594
A3	297	420
A4	210	297
A5	148	210

Figure 3.1: Water plants

Chapter 4

Conclusions

Appendix A

Mathematical proofs

A.1 Euler's equation

Euler's equation gives the relationship between the trigonometric functions and the complex exponential function.

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{A.1}$$

Inserting $\theta = \pi$ in (A.1) results in Euler's identity

$$e^{i\pi} + 1 = 0 \tag{A.2}$$

A.2 Navier Stokes equation

The Navier–Stokes equations mathematically express momentum balance and conservation of mass for Newtonian fluids. Navier-Stokes equations using tensor notation:

$$\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x_j}\left[\rho u_j\right] = 0 \tag{A.3a}$$

$$\frac{\partial}{\partial t}\left(\rho u_i\right) + \frac{\partial}{\partial x_j}\left[\rho u_i u_j + p\delta_{ij} - \tau_{ji}\right] = 0, \quad i = 1, 2, 3 \tag{A.3b}$$

$$\frac{\partial}{\partial t}\left(\rho e_0\right) + \frac{\partial}{\partial x_j}\left[\rho u_j e_0 + u_j p + q_j - u_i\tau_{ij}\right] = 0 \tag{A.3c}$$

Appendix B

Experimental results

List of references

Cundall, P.A. and Strack, O.D.L. (1979). A discrete numerical model for granular assemblies. *Géotechnique*, vol. 29, no. 1, pp. 47–65.

Lin, X. and Ng, T.T. (1997). A three-dimensional discrete element model using arrays of ellipsoids. *Géotechnique*, vol. 47, no. 2, pp. 319–329.